SHAGGAR INSTITUTE OF TECHNOLOGY
Excellence in Science and Technology

Math 1120: Calculus II - Course Syllabus

1. Basic Course Information

- Course Code and Title: *Math 1220: Calculus II*
- Credit Hours: *4 crs*
- Instructor(s): *Prof. Abdulkadir Hassen, Prof. Aklilu Zeleke, Dr. Fikreab Solomon, Dr. Fufa Beyene, and Mr. Lemessa Berewak*
- Semester/Quarter: *Winter 2026, Second Term*
- Prerequisites: *Math 1110: Calculus I*
- Co-requisites: *None*
- Class Schedule: *Monday, Wednesday, and Friday 2:00pm to 3:15pm.*
- Office Hours: TBA
- Contact Information: *ahassen09@gmail.com* and *zeleke@msu.edu* and *fabmaysolj@gmail.com* <fbbeyenefufa1@gmail.com;lemessa4berewak@gmail.com;

2. Course Description

This course begins with a review of applications of definite integrals, and covers the following topics: techniques of integration, numerical methods for integrals, improper integrals, infinite series: geometric and p-series, tests of convergence, power and Taylor series for some elementary functions, parametric equations and polar coordinates, vectors in 2- and 3-spaces, dot and cross products. Emphasis will be given to the applications of these concepts in areas of physical, biological, and engineering.

3. Course Objectives

Students will demonstrate the ability to:

1. *Set up integrals to find volumes of regions whose cross-section areas can be found*
2. *Use integration to find areas and volume*
3. *Use integration to find work done by a force and average values of a function*
4. *Perform integration by parts, partial fractions and various substitutions as well as selected numerical techniques*
5. *Recognize and evaluate improper integrals*
6. *Determine convergence and divergence of infinite series*
7. *Find Taylor series and their radius and interval of convergence*
8. *Work on group project for deeper understanding of derivative and integration*
9. *Graph and find areas in polar coordinates*
10. *Calculate dot and cross product*
11. *Identify and find equations for lines, planes and quadric surfaces,*

4. Student Learning Outcomes (ABET Outcomes 1–7)

Upon successful completion of the course, students will be able to:
1. Identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.

2. To produce solutions that meet specified needs with consideration for public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.

3. Communicate effectively with a range of audiences.

4. Recognize ethical and professional responsibilities in engineering situations and make informed judgments.

5. Function effectively on a team to provide leadership, create an inclusive environment, establish goals, plan tasks, and meet objectives.

6. Develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.

7. Acquire and apply new knowledge as needed, using appropriate learning strategies.

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5. Mapping to Program Outcomes

ABET Outcomes	Course Objectives
1. Identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.	1, 2, 3, 9
2. To produce solutions that meet specified needs with consideration for public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.	11
3. Communicate effectively with a range of audiences.	11
4. Recognize ethical and professional responsibilities in engineering situations and make informed judgments.	
5. Function effectively on a team to provide leadership, create an inclusive environment, establish goals, plan tasks, and meet objectives.	11
6. Develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.	
7. Acquire and apply new knowledge as needed, using appropriate learning strategies.	1-11

6. Required Textbooks and References

Text: Calculus: Early Transcendentals, 9th Edition, by James Stewart, Daniel K. Clegg, Saleem Watson.

References:

1) *Calculus, Early Transcendental, 4th Edition, by Jon Rogawski, Collin Adams, and Robert Franzosa.*

2) *Calculus, Volume I1:* https://openstax.org/books/calculus-volume-2/pages/1-introduction

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7. Course Topics and Schedule

Content: (12 weeks 4 lectures per week = 48 lectures) In this course, we will cover Chapters 7 through 11 of the textbook by James Stewart et al. listed above. The schedule below shows the weekly topics along with the corresponding exams.

Week(s)	Topic	Assignments
1, 2	**Ch 6 and 8. Applications of Integral** 6.1. Area Between Two Curves 6.2. Volumes 6.3 Volumes by Cylindrical Shells 6.4 Average Value of a Function, Work, and Energy 8.1 Arc Length 8.2 Area of a Surface of Revolution 8.3 Applications to Physics and Engineering	HW 1, 2 Q1(Quiz 1) **Test 1 (Ch 6, 8)**
3 and 4	**Chapter 7 Techniques of Integration** 7.1 Integration by Parts 7.2 Trigonometric Integral 7.3 Trigonometric Substitution 7.4 Integration by Partial Fractions 7.7 Approximate Integration 7.8 Improper Integral	HW 3 and 4 Q 2
5 and 6	**Chapter 10. Parametric Equations and Polar Coordinates** 10.1 Curves Defined by Parametric Equations 10.2 Calculus with Parametric Curves 10.3 Polar Coordinates 10.4 Calculus in Polar Coordinates	HW 5, 6 Q 3 **Test 2 (Ch 7, 10)**
6, 7, and 8	**Chapter 11. Sequences, Series, and Power Series** 11.1 Sequences 11.2 Series 11.3 The Integral Test and Estimates of Sums 11.4 The Comparison Tests 11.5 Alternating Series and Absolute Convergence 11.6 The Ratio and Root Tests 11.7 Strategy for Testing Series 11.8 Power Series 11.9 Representations of Functions as Power Series 11.10 Taylor and Maclaurin Serie	HW 7, 8, and 9 Q 4 **Test 3 (Ch 11)**

9 and 10	**Chapter 12. Vectors and the Geometry of Space** 12.1Three-Dimensional Coordinate Systems 12.2 Vectors 12.3 The Dot Product 12.4 The Cross Product	HW 5, 6 Q 3 **Test 2 (Ch 7, 10)**
11	Review, presentation, and Final Exam	***Final Exam (All Chapters)***

8. Laboratory Component

Mathematica or MATLAB

9. Grading Policy

Your final grades will be determined by your results on weekly quizzes, two tests, final exam, projects, homework assignments, and class participation. The dates for each test will be announced in class one week before the test date. The materials covered in the for the m tests will be as follows:

- **Test 1:**(15%) covers Chapters 6 and 7
- **Test 2:** (15%) covers Chapter 8 and 10
- **Test 3:** (15%) covers Chapter 11.
- **Quizzes**:(15%)
- **Homework**:(10%)
- **Projects:** (10%)
- **Final Exam:** (20%) covers all chapters

The letter grade assignment will be determined at the end of the term.

Homework: The homework problems are posted on the course page of the SIT's LMS.
Projects: These can be found on SIT's LMS page of the course. Due dates will be announced in advance.

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Excellence in Science and Technology

For SIT's policies about Academic Integrity, Class Attendance, Classroom Behavior, Accessibility and Student Support, and Safety Considerations, please read the SIT Student Manual.

In particular the following three policies are essential for this course.

Academic Honesty: *Cheating on a test or assignment seriously undermines the integrity of the academic system and will not be tolerated. Although a student is not cheating, he or she is expected to refrain from actions that could be suspicious. Using common sense on your part should avoid unnecessary embarrassment.*

Attendance: *Attendance will be taken at every meeting. Students are expected to attend every online class and to be on time. If you are absent from any online and/or in person class for any reason, it is your responsibility to find out what you missed, including any announcements.*

Classroom Behavior: *You are expected to be in the classroom before the start of class time and leave at the end of the class time. Excessive talks without the permission of the instructors is prohibited. The use of any electronic device is allowed for the lecture and class activities only.*